AYANANTA CHOWDHURY

AI-Native Documentation Architect | Senior Manager, Product Documentation

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Professional Summary

I am an AI-native Help Architect with **15+ years** of experience in enterprise software documentation, structured authoring, semantic content engineering, and documentation governance. I specialize in content strategy, terminology governance, taxonomy design, DITA optimization, and AI-driven content architectures. I build semantic content maturity models by:

- Normalizing existing documentation with controlled vocabulary, using an industry-aligned **terminology** framework.
- Generating Schematron **enforcement rules** to validate appropriate terminology usage.
- Designing a multi-faceted **content taxonomy** using Carrot^2.
- Chunking content into **atomic units**—each with a unique identifier and tagging them with taxonomy facets.
- Enhancing documentation with iiRDS-compliant **semantic metadata**.
- Building formal semantic models using OWL, RDF, RDFS, and SKOS to develop **content ontologies**.
- Enabling **knowledge graph** construction and hybrid retrieval (**vector + graph RAG**).

Work Experience

HCL Technologies | Senior Technical Manager – User Assistance (Mar 2023 – Apr 2025, Remote) Fulltime

- Audited content by running TEXTML queries on IxiaSoft CCMS to extract exact, near, and thematic duplicates from the documentation of a SaaS ERP solution from SAP. Applied content reuse strategy to optimize content by 11%.
- Extracted product/domain terms and phrases using SpaCy. Determined their canonical forms and created a glossary. Recorded the canonical (preferred) terms, as well as their allowed and deprecated variants. Defined the terms in coordination with the team. Implemented term keyrefs to normalize documentation with controlled vocabulary.
- Enforced content governance by implementing Schematron rules to ensure consistency and compliance with goals.
- Chunked content to indicate their taxonomy facets via semantic metadata.
- Built an ontology by defining key concepts (classes), hierarchical structure, properties (attributes), and relationships.

Quincus | API Technical Writer (Oct 2022 – Nov 2022, Remote) Fulltime

- Worked toward improving the developers' API usage experience by explaining the available API endpoints, resources, methods, authentication protocols, parameters, and headers (logistics domain).

SugarBox Networks | **Technical Communications Manager** **(Aug 2020 – Sept 2022, Remote)** **Fulltime**

- Managed content backlogs in JIRA, delegated doc tasks, reviewed documentation, and assessed team's performance.
- Built a controlled vocabulary by canonicalizing domain terms into a glossary and normalizing content via term keyrefs.
- Created a taxonomy, chunked content into atomic units for semantic tagging to explicitly reflect their taxonomy facets.
- Measured product documentation success using Google Analytics, Hotjar, and Algolia.
- Created user personas and user workflow diagrams to understand customer needs and produce tailored content.

HCL Software | **Technical Writing Senior Consultant** **(Nov 2018 – Mar 2020, Pune)** **Fulltime**

- Led and mentored the team in the documentation effort for a data engineering product, following CI/CD workflow.
- Drafted Hotfix docs and release notes for a data-engineering solution and wrote a variety of UI texts (microcopy).
- Added parameter descriptions in the Open API Specification file and wrote the conceptual part of API documentation.

Sungard Discovery Services | **Specialist, Technical Documentation** **(May 2017 - Jul 2017, Pune)** **Fulltime**

- Produced video tutorials for the company's LMS platform, explaining the capabilities of a disaster recovery product.

Persistent Systems | **Information Development Team Lead** **(Nov 2013 – May 2017, Pune)** **Fulltime**

- Produced demo videos of medical device MVPs for potential investors (life sciences & healthcare domain).
- Coordinated with cross-functional teams of Merck & Co. for ITIL-based change management, using ServiceNow.
- Collaborated with the VMware on-site team to author the user and quick start guide for vRealize Network Insight.
- Ensured content followed WCAG 2.1 accessibility standards. Did usability testing to verify content accuracy.

Rolta India | **Senior Technical Writer - Product Documentation** **(Aug 2012 – Jul 2013, Mumbai)** **Fulltime**

- Authored content to explain new features developed for a GIS product using a single-sourcing tool, following Agile

Polaris Software Lab | **Technical Writer** **(Oct 2010 – Aug 2012, Mumbai)** **Fulltime**

- Co-authored user and installation guides for a banking solution processing cross-border transactions via SWIFT messaging, complying with NACHA standards for ACH transfers, and ensuring accurate fund settlements.

Connectiva Systems | **Technical Writer** **(Feb 2008 – Sept 2010, Kolkata)** **Fulltime**

- Built storyboards for eLearning videos (telecom domain), edited corporate training PPTs, and updated the user guide.

Education

Postgraduate in Mass Communication and Journalism, Calcutta University | 2005–2006

Certifications

DITA XML Authoring | UX Writing | API Documentation | Camtasia | Git

Achievements

- Increased 11% content reuse by implementing “conref” & “conkeyref” throughout an SAP ERP solution documentation.
- Reduced CDN configuration time by 31.4% (measured via Hotjar) at SugarBox Networks with tailored documentation.

Skills / Competencies

AI

Design hybrid AI architectures combining LLM-based generation with neuro-symbolic knowledge graphs to power proactive, agentic assistance. Apply prompt engineering techniques to optimize generative AI responses for daily tasks.

INFORMATION ARCHITECTURE

Build a terminology framework based on ISO/IEC/IEEE 24765, define authoring-time terminology validation rules, construct a content taxonomy model, perform data modelling, implement citation guidelines, and enrich content with metadata. Create formal semantic models using OWL, RDF, RDFS, and SKOS to build an ontology. Proficient in Carrot^2 Protégé; have familiarity with PoolParty. Build, manage, and scale knowledge graphs using Neo4j to define entities, relationships, constraints, & reasoning rules. As a traditional information architect, I optimize documentation for scalability.

CONTENT STRATEGY

Define user personas, map user journeys, and determine the appropriate help type and channels for distribution based on audience analysis. Create content calendars & implement performance measurement mechanisms to assess performance.

CONTENT DEVELOPMENT

Create multimodal help materials that guide users in setting up, operating, and maintaining enterprise applications. The help materials consist of video tutorials, screen recordings, printed and online guides, as well as data visualizations such as GIFs, infographics, diagrams, flowcharts, charts, and mind maps. Author release notes, what’s news, hotfix docs, FAQs & troubleshooting, training materials for certificate exams/corporate training/eBooks. Write SEO-optimized (E-E-A-T) marketing collaterals, such as blogs, newsletters, and case studies. Edit SOPs & runbooks as per MSTP guidelines. Edit technical specifications and functional documents and create process flow documentation (business scenario explorers).

UX WRITING (UI TEXTS) - Embed assistance in the form of microcopy and walkthrough videos (digital guided tours).

API DOCUMENTATION

Deliver exceptional developer experiences through well-structured API documentation with interactive examples.

LANGUAGE: JSON, XML, YAML, Markdown, HTML (working knowledge), Python (only familiarity)

Tools

Component Content Management System: IxiaSoft | **Docs-as-Code**: Git, Markdown | **Audio-Visuals & Diagramming & Flowcharting**: Camtasia, Visio, Napkin, Piktochart, HeyGen, and Lucidchart | **Analytics**: Google Analytics, Hotjar, and Algolia | **Authoring**: DITA XML Editors (Oxygen, Arbortext), RoboHelp / MadCap Flare, Confluence, Docs, MS Office | **DAP**: WalkMe | Project Management/Ticketing ITSM: JIRA, ServiceNow | **API Documentation**: Swagger, Postman | **Vibe Coding**: Cursor AI